

SC4001: Neural Networks and Deep Learning

Programming Assignment

Part A: Regression with DNN

Predict Singapore **housing price** from both **categorical** and **continuous** inputs

Question A1 (baseline model)

- Each categorical feature is first indexed and then converted to an embedding of dimension 8 by using **nn.Embedding** layer consisting of learnable weights.
- Draw a **block diagram** of the baseline network; compute the number of learnable parameters
- Find the optimal parameters: **embedding dimension** x **hidden-layer width** by using **ray[tune]**

Question A2 (ablation)

- (i) Remove categorical variables
- (ii) Replace ReLU layer with sigmoidal layer
- Perform experiments on **five seeds** (`torch.manual_seed(seed)`), report mean and s.d., and compare with the baseline

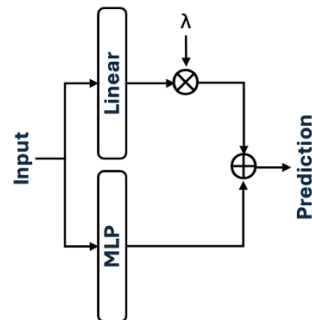
Part A

Question A3 (wide-and-deep architecture)

- output = MLP output + $\lambda \times$ linear output

Find the optimal parameter $\lambda \in \{0, 0.25, 0.5, 1.2\}$

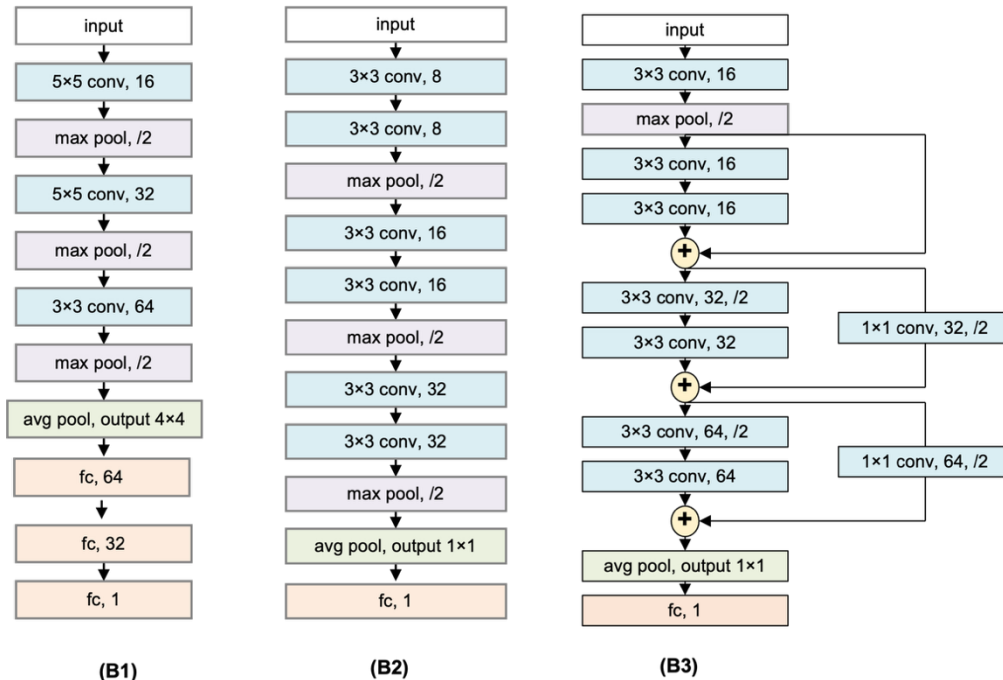
- Evaluate the optimal network over 5 seeds and compare with the baseline architecture



Question A4 (XAI)

- Use **FeatureAblation** method on **Captum** library
- Find feature **importance scores** in Singapore dollars

Part B: classification with CNN



(B1) **AlexNet** like architecture, (B2) **VGG** like architecture, and (B3). **ResNet** like architecture.

All the layers use **ReLU** activation function including after residual addition $\oplus$. All convolutional layers use **BatchNorm** and **SAME padding**. Layers indicating /2 have a **stride** of two and all other layers have stride of 1. **Dropout** = 0.5 is applied only in the hidden fully-connected (fc) layers in B1 architecture.

Part B

Question B1 (AlexNet-like architecture)

- Find the **optimal learning rate** from $\{10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}\}$
- Plot learning curves, evaluate, etc.

Question B2 (VGG-like architecture)

- Find the **batch size** from $\{16, 32, 64, 128, 258\}$
- Plot learning curves, evaluate, etc.

Question B3 (ResNet-like architecture)

- Use learning rate from B1 and batch size from B2; Plot learning curves, evaluate, etc.
- Compare B1-B3 architectures: performances, number of learnable parameters, FLOPs, etc.

Question B4 (XAI)

- Use **Grad-CAM** method on **Captum** library
- **Overlay** Grad-CAM heat maps on X-ray images

Housekeeping rules

- Assessment is based on **the report** (6 pages each). Codes will be used for checking.
PartA_report_YourName.pdf, PartA_codes_YourName.zip
PartB_report_YourName.pdf, PartB_codes_YourName.zip
- Marks: **45** for Part A + **45** for Part B + **10** for **presentation**. Follow the instructions closely. Any deviations may lead to loss of marks for presentation.
- Codes must be in .py and .ipynb, tidied, properly labelled, and ready to run. Include a **README** file.
- **Generative AI declaration**
- Late submissions: **penalized** for 5 marks for each day up to 3 days.
- This assignment is to be done **individually**. Absolutely NO copying, duplicating, or plagiarism.
- Post your queries on the **Discussion Board** in NTULearn
- Specific questions for help via cx4042@e.ntu.edu.sg
- TA: Deepank Girish, Yan Yige, Tiara Natasha, and Ajith Senthisenan.